

Who verifies the number? Market concentration and verifier effects in mandatory emissions verification

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Abstract

Third-party verification turns a mandatory emissions registry into the number that an emissions trading system prices, yet the market that supplies verification is rarely observed and regulators have few tested instruments for governing it. Using the complete Korean mandatory greenhouse-gas registry (K-ETS and Target Management Scheme statements for 2011–2024; 12,729 entity-years, 1,848 entities, 24 verifiers), we ask two policy questions: should regulators worry about concentration in the verification market, and does the choice of verifier change the certified number? The market concentrated between 2013 and 2024: the client-count Herfindahl index rose from 1,274 to 1,777. Concentration came from new entrants' first choice of verifier (+837 clients for the three largest verifiers) and from the en-bloc movement of client books after six designation cancellations, recorded without a stated cause by the National Institute of Environmental Research; voluntary switching by incumbents dispersed clients (−61). Within entities that switched verifier, with entity and industry-by-year fixed effects on post-conformity-assessment certified values, verifier identity leaves no detectable trace on certified intensity where the monitoring rules fix emission factors (entities below 500

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ktCO₂): 10 of 16 verifier effects pass a $\pm 5\%$ equivalence test and no major verifier's 95% confidence interval extends beyond $\pm 7.1\%$. When the largest cancelled verifier's clients moved en bloc to a single successor, their certified intensity did not move detectably (95% confidence interval within about $\pm 7\%$). Where the rules leave discretion, the data are uninformative below a median minimum detectable effect of 19.7%. A joint signal in certified emission and energy levels does not survive multiple-testing correction and disappears once entities whose energy use jumped at the switch are excluded; a low-powered test of the regulator's verifier-evaluation disclosures is borderline. The evidence is consistent with factor-fixing monitoring rules, rather than the choice of verifier, pinning certified intensity in those industries, and it points to the entry stage and to what happens to client books after a cancellation as the two channels through which concentration arose.

Key policy insights

- Korea's verification market concentrated (Herfindahl index 1,274 to 1,777, 2013–2024) through new entrants' first choice of verifier and the en-bloc movement of client books after designation cancellations, not through incumbents switching; policy on market structure must act at those two points.
- Where monitoring rules fix emission factors, no major verifier's 95% confidence interval for certified intensity extends beyond $\pm 7.1\%$ (10 of 16 pass $\pm 5\%$ equivalence); supervision need not target verifier choice there.
- When a large cancelled verifier's clients moved en bloc to one successor, certified intensity did not change detectably (interval about $\pm 7\%$); cancellation reallocated the market, not the certified number.

Keywords: emissions trading; monitoring, reporting and verification (MRV); third-party verification; regulatory intermediaries; market concentration; equivalence testing; Republic of Korea; K-ETS

1 Introduction

An emissions trading system prices a number that a third party has certified. Every compliance obligation, every allocation and every penalty rests on the verified emissions statement, so the credibility of the system is the credibility of its verification. Two questions follow for the regulator that designates verifiers. Should it worry about concentration in the market that supplies verification, and if so, at which point does concentration arise: when a newly covered entity chooses its first verifier, when a cancelled verifier’s clients move, or when incumbents re-select? And does the identity of the verifier change the certified number? If it does not, verifier rankings and capability assessments are instruments of market structure rather than of measurement accuracy; if it does, the segment of the rules in which it does is where supervision belongs.

These questions have been hard to answer because verification markets are rarely observed. The literature on third-party verification in carbon markets consists of normative and legal analyses of the verifier’s incentives (Lund, 2010; Xie and Faure, 2026), diagnoses of verifier capacity at the launch of a system (Tang et al., 2018; Zeng et al., 2018) and comparisons of self-reported with verified emissions in the first years of pilot markets (Zhang et al., 2019). The structure of the verification market itself has been discussed by analogy with the audit market for financial statements (U.S. General Accounting Office, 2003; U.S. Government Accountability Office, 2008; Kohlbeck et al., 2008; DeFond and Lennox, 2011). Regulators have nonetheless acted: since 2021 the EU has required the lead auditor of an installation to rotate after five consecutive verifications, a measure aimed squarely at the impartiality risk of a repeated engagement. We have not found a peer-reviewed study that follows a mandatory emissions verification market for more than a decade, decomposes the channels of its concentration, bounds the effect of verifier identity on certified values within entities, and tests what happens to certified values when a cancelled verifier’s clients are absorbed by a successor.

Korea’s mandatory registry allows all three. The annual key-information workbooks of

entity-level emissions statements list, for every entity under the Korean Emissions Trading Scheme (K-ETS) and the Target Management Scheme (TMS), the certified emissions, energy use and the verifier of record. Assembled for 2011–2024, they give 12,729 entity-years, 1,848 entities and 24 verifiers. Two features of the Korean rules make the second question answerable. The monitoring guideline fixes, by activity and facility size, the minimum calculation tier: for combustion below 500 ktCO₂ and for purchased electricity and heat, the emission factor is a default or national coefficient, so certified intensity (tCO₂ per TJ) is pinned by the rules and the verifier has no factor to accept or reject. In process and waste activities and in the largest facilities, plant-specific parameters are allowed and the verifier’s judgement matters. The rules therefore define, in advance, a segment in which verifier identity should leave no trace and a segment in which it could.

Our answers are three. First, the market concentrated: the client-count Herfindahl index rose from 1,274 in 2013 to 1,777 in 2024 and the number of active verifiers fell from 21 to 13. The concentration came from new entrants, who chose the three largest verifiers more often than incumbents in every year, and from the absorption of cancelled verifiers’ client books by single successors; voluntary switching by incumbents moved clients away from the largest verifiers. Second, in the factor-fixed segment the identity of the verifier leaves no detectable trace on certified intensity: the joint test does not reject, 10 of 16 verifier effects pass a $\pm 5\%$ equivalence test, and no major verifier’s 95% confidence interval extends beyond ± 7.1 log points. In the discretion segment the data cannot detect effects below roughly 20%. Third, when the largest cancelled verifier’s 71 clients moved to one successor, their certified intensity did not move detectably.

We read these results through the regulator–intermediary–target model of Abbott et al. (2017): verifiers are the intermediaries through which the regulator reaches emitters, and the regulator’s instruments on the intermediary—the tier rules, designation and its cancellation, and public evaluation—map onto our three results. The paper adds to the literature on MRV design (Bellassen et al., 2015; Shishlov and Bellassen, 2016; Warnecke, 2014) and

on K-ETS (Oh et al., 2017; Howie et al., 2020; Jeong and Chang, 2025) a first quantitative account of a mandatory verification market. The analysis has three layers, kept apart throughout: descriptive market structure, a bounded null on verifier effects estimated under an analysis plan frozen before confirmatory estimation, and exploratory results reported in the Supplementary Material.

2 How three regulators govern verification

Table 1 sets the instruments through which Korea’s regulator governs verification beside those of the EU ETS and of China’s national ETS, each cell read from the governing legal text. The verified entity selects and pays its verifier and attaches an impartiality self-assessment to every contract; independence rules operate at the level of the firm (consulting and abatement-service firms cannot be designated; a verifier may not advise the entity it verifies) and of the person (a two-year cooling-off for former employees and consultants); designation is statutory, valid for three years and, since June 2021, administered by the National Institute of Environmental Research (NIER), with a separate ISO/IEC 17029 and ISO 14065 accreditation track created in September 2021 for international mutual recognition.

Three of the comparisons matter for how the results should be read. First, who chooses: Korea and the EU both leave the choice and the fee to the emitter, a structure whose incentive problem has been the standard criticism of third-party verification since Lund (2010), while in China the provincial authority is itself the verifying body and procures technical service institutions by tender, so the market we study does not exist there in the same form. Second, rotation: since 2021 the EU has required the lead auditor to step away from an installation after five consecutive verifications, the first rotations falling due after the 2025 data year, whereas Korea caps nothing at either the firm or the auditor level and China has no rotation rule either. Korea is therefore a setting in which the impartiality risk that the EU rule targets is left unmanaged, which is what makes a bounded null on

verifier identity informative. Third, disclosure: China requires each province to evaluate and publish the quality, compliance and timeliness of every verification body every year, the EU publishes no performance ranking at all, and Korea published two rounds in 2019 and 2020 and then stopped. A fourth comparison frames the last result: neither Korea nor China has a published procedure for what becomes of a cancelled verifier’s clients, and the EU rule simply requires the operator to hire another verifier, with no transitional provision.

Three instruments matter for what follows. The first is the tier rule of Annex 5 of the MRV guideline, which fixes the minimum calculation tier by activity and facility size (Tier 1 below 50 kt, Tier 2 between 50 and 500 kt, Tier 3 at 500 kt and above for stationary combustion; national grid factors for purchased electricity at every size). The TMS applies the same annex by reference. The second is the regulator’s own conformity assessment of every statement after verification (3,586 errors were found in the 2018 statements according to the 2019 release), which means that the certified values we observe are post-assessment values. The third is public evaluation of verifiers: the Ministry of Environment published per-verifier error rates, contribution rates and correction rates for 17 verifiers in August 2019 and a composite 100-point score for 16 verifiers in October 2020; no further public round was found for 2021–2024 despite an announced annual cycle.

Six verifiers that were active in the registry had their designation cancelled between February 2018 and October 2022. NIER’s register records the dates; the cause column is blank for all six. Two of the 2022 cancellations coincide with the first accreditation round and none of the three verifiers cancelled in 2022 is among the 13 accredited, but no public record links the two events and we do not infer a cause. No public record describes how the client books of a cancelled verifier are handled.

Table 1: Instruments through which regulators govern third-party verification of emissions: Korea, the EU and China

Instrument	Korea (K-ETS and Target Management Scheme)	EU ETS	China national ETS
Who selects and pays the verifier	Entity contracts and pays; an impartiality self-assessment is attached to every contract.	Operator contracts and pays; guidance treats this as a source-of-revenue risk to impartiality.	Provincial authority verifies and procures the institutions through government purchase of services; the emitter pays only for testing and report preparation.
Verifier independence rules	No consulting firm may be designated; no advice to the client verified; 2-year cooling-off for auditors and former employees.	Consulting or technical assistance on monitoring for the same client is an unacceptable impartiality risk; a repeat engagement requires a documented risk assessment.	No consulting, carbon-asset management or ownership ties; 3-year cooling-off for shared managers and for auditors who advised the emitter; report preparation and technical review may not be combined within a province.
Mandatory verifier rotation	None, at either the body or the auditor level.	None for the body; the lead auditor must take a 3-year break after 5 consecutive verifications of the same installation, first applying after 2025 data.	None; instead the on-site visit may be waived after two years without a non-conformity.
Designation or accreditation	NIER designation, 3-year validity, annual assessment; a separate ISO 17029 and 14065 accreditation track since 2021.	Accreditation by each Member State's single national accreditation body against EN ISO 14065; at most 5 years, with annual surveillance, reassessment and peer evaluation of the bodies.	No qualification approval; provinces select by tender. A 2025 opinion directs the introduction of qualification management with entry and exit rules.
Minimum tier rules that fix factors	Annex 5: Tier 1 below 50 kt, Tier 2 at 50-500 kt, Tier 3 at or above 500 kt for combustion; national factor for purchased electricity at every size.	Categories A (at most 50 kt), B and C; category A and commercial standard fuels use minimum tiers, all others the highest tier, relaxable where infeasible or costly.	No size-based tiers; the carbon content of coal must be laboratory-tested monthly or a conservative default applies.
Regulator's own conformity assessment of certified values	KECO conformity assessment after verification (3,586 errors in the 2018 statements); registry values are post-assessment.	The competent authority estimates emissions conservatively where verification is absent or non-compliant, and reports verifier non-compliance to the accreditation body each year.	Verification is the authority's own act; results are published within 7 working days and monthly data pass a three-tier review (over 3 million power-sector records in 2024).
Public evaluation of verifiers	Two rounds only (2019, 2020); none found for 2021-2024 despite an announced annual cycle.	None. Only a public database of accredited verifiers, their scope and any administrative measures imposed on them.	Annual and mandatory: each province evaluates and publishes every institution's quality, compliance and timeliness (16,600 records in 2023, 99.7 per cent qualified).
Suspension and cancellation of designation	The register lists dates; 6 active verifiers cancelled 2018-2022, with the cause column blank for all.	The accreditation body must suspend for a serious breach and withdraw for fraud or false information; measures are public and notified across Member States (one withdrawal EU-wide in 2020).	No accreditation to suspend; fines of 5 to 10 times illegal gains or RMB 0.2-1m, a ban from the business in serious cases, and penalties entered in the public credit record.
Procedure for client books after a cancellation	None in public records; the observed pattern is absorption by a single successor (87, 100 and 81 per cent).	The operator must hire another verifier; there is no transitional rule. Without compliant verification by 31 March, emissions are estimated conservatively and transfers blocked.	No explicit rule; the provincial authority reallocates the tasks it has procured.

Sources: Korea: verification guideline (Ministry of Environment Notice 2024-169 and its 2018 and 2021 predecessors), MRV guideline Annex 5 (Notice 2025-64), NIER Notice 2021-65, the NIER register of designations, suspensions and cancellations (2024-07-01 and 2025-01-06 issues), Ministry of Environment releases of 22 August 2019 and 27 October 2020, and the registry transitions computed in this paper. EU: the consolidated Accreditation and Verification Regulation (EU) 2018/2067, the Monitoring and Reporting Regulation (EU) 2018/2066, Directive 2003/87/EC, Regulation (EC) 765/2008, and Commission guidance and FAQ documents. China: the Interim Regulations on Carbon Emission Trading (State Council Order 775, 2024), the Administrative Measures (MEE Order 19, 2021), the Verification Guideline (MEE 2021-130), the power-sector accounting guideline (MEE 2022-485), the annual MEE work notices and the National Carbon Market Report. Supplementary Material S4 gives the full text of every cell with the article it was read from, and the reproduction package holds the URLs.

3 Data and methods

3.1 Data

The panel is assembled from the annual key-information workbooks of entity-level emissions statements for 2011–2024, which list for every K-ETS and TMS entity the supervising ministry, designation type (company or site), designated industry, certified emissions (tCO₂e), energy use (TJ) and the verifier of record. Verifier names are harmonised to 24 codes (Supplementary Material S1). Two rows without a verifier are dropped, leaving 12,729 entity-years. Rows with non-positive or missing energy (27) or emissions (4) are excluded from log outcomes only.

The 2012 verifier column duplicates the 2011 column for 99.1% of entities, so every series that depends on changes starts in 2013; 2011–2012 appear only in grey in Figure 1. The designated-industry field is complete through 2019 but sparse from 2020 (88% in 2020, 53% in 2021, 33–36% in 2022–2024); we bridge it by carrying each entity’s last designated industry forward and, for new entities, by a crosswalk from the planning industry, reaching full coverage with 77 industry labels (Supplementary Material S1 records the source of every label).

Consecutive entity-year pairs (10,823) carry the switch and its type. Of 1,102 switches in 2013–2024 (an annual rate of 10.6%), 944 are voluntary, 110 are cancellation-linked block transfers (the client of a verifier that disappears from the registry reappears under the verifier that received at least half of that book), 36 are cancellation-linked dispersed transfers and 12 are pilot-period dropouts. Pairs also carry flags for changes in designation type, industry or ETS status; ‘scope-stable’ pairs have none. The verifier-effect estimates use the 613 entities observed with at least two verifiers, which are observed longer (8.9 versus 5.2 years), are more often ETS entities (55% versus 39%) and have more variable intensity than the 1,214 never-movers (Supplementary Material S2).

The fixed-factor group is defined from Annex 5: entity-years whose designated industry is

in a combustion, mobile, building or energy activity class and whose emissions are below 500 kt. It holds 47% of entity-years in 2013–2024 but 5.4% of certified emissions; the discretion group holds the rest.

3.2 Methods

Market structure. We compute, by registry year, the number of active verifiers, the Herfindahl index (HHI) of client counts and of certified emissions, and the client share of the three largest verifiers (CR3). Changes in the client count of the three largest verifiers of 2024 are decomposed exactly into six channels: entrants’ first choice, entities leaving the registry, and the net flows through voluntary, block, dispersed and pilot-period switches; the same decomposition is expressed in share points by separating numerator channels from the market-growth (denominator) effect. Two counterfactual HHI paths reassign every cancellation-linked switch to the surviving verifiers in proportion to their prior-year shares or equally, holding those clients there until their next voluntary switch; because the answer depends on the rule, we report both.

Verifier effects. For outcome y in $\{\ln(\text{tCO}_2/\text{TJ}), \ln \text{tCO}_2, \ln \text{TJ}\}$ we estimate $y_{it} = \alpha_i + \gamma_{j(i)t} + \sum_v \beta_v \mathbf{1}[\text{verifier}_{it} = v] + e_{it}$ with entity fixed effects α_i and industry-by-year fixed effects γ_{jt} , by alternating projections (Guimarães and Portugal, 2010), on movers only and with the largest verifier (V1) as reference. The covariance is clustered by entity with the usual small-sample correction; each β_v is accompanied by its 95% confidence interval, the number of switches that identify it, the minimum detectable effect at 80% power (MDE = 2.8 standard errors) and a two-one-sided equivalence test at $\pm 2\%$ and $\pm 5\%$ (Lakens, 2017), declared passed when the 90% confidence interval lies inside the margin. A joint Wald test is accompanied by a wild cluster bootstrap p -value (Cameron et al., 2008), with Rademacher weights over entities and 999 draws for individual coefficients. The bound we report is the largest absolute 95% confidence limit among the seven largest verifiers; point-

estimate dispersion is never used as a bound. A placebo keeps each entity’s switch timing and redraws the destination verifier from industry-by-year shares (200 draws). An identity check regresses $\ln tCO_2$ on $\ln TJ$ within the fixed-factor group. Holm’s adjustment (Holm, 1979) is applied across the 24 joint tests reported in the paper and its supplement. Coefficients on log outcomes are reported as log points $\times 100$.

Cancellation-linked transfers. For the three block transfers with at least 13 treated entities we estimate an event-study difference-in-differences over ± 3 years with the same fixed effects, treated entities being clients of the cancelled verifier that appear under the successor in the transfer year, and two control groups (the successor’s continuing clients; all non-switchers of that year). We report the joint test on the leads, the average post-transfer coefficient with its bootstrap p -value and its equivalence test.

Discipline. The sample rules, the fixed-factor split, the inference rules, the bound definition, the reading rules for the event study and the hypotheses of the supplementary tests were written into an analysis plan and frozen with SHA-256 hashes on 19 September 2026, after feasibility checks and before confirmatory estimation on the same day; all reported estimates are post-freeze estimates, and every departure is logged (Supplementary Material S11). Two caveats apply throughout: the certified values have passed the regulator’s conformity assessment, so detected errors were removed before we observe them; and verifier effects are identified only within entities that switched.

4 Results

4.1 A concentrating market, 2013–2024

Figure 1 and Panel A of Table 2 describe the market. The client-count HHI rose from 1,274 in 2013 to 1,777 in 2024, with a peak of 1,911 in 2022 and a fall to 1,780 in 2023;

the emissions-weighted HHI rose from 1,693 to 2,121; the client share of the three largest verifiers rose from 0.536 to 0.672 (0.706 in 2022); the number of active verifiers fell from 21 to 13. Fifteen of the 24 verifiers in the 2011–2012 pilot period were still active in 2013; two pilot-period dropouts left in 2013 and two in 2014; six designation cancellations followed between 2018 and 2022.

Where did the concentration come from? Panel B decomposes the change in the client count of the three verifiers that were largest in 2024, from 300 of 599 entities in 2013 to 782 of 1,164 in 2024. Entrants’ first choice added 837 clients to the three, entities leaving the registry removed 419, voluntary switches by incumbents removed a net 61, cancellation-linked block transfers added 98 and dispersed transfers 24. In share points, net of the growth of the market, entrants raised the fixed top-3 share by 0.191, leavers lowered it by 0.075, voluntary switches lowered it by 0.052 and block transfers raised it by 0.084; the total change was 0.171 (0.501 to 0.672). Entrants chose one of the three largest verifiers more often than incumbents in every year (a first-choice share of 0.578–0.815 against 0.491–0.697 for incumbents), and the HHI of entrants’ choices exceeded the incumbents’ HHI in every year (1,750–3,062 against 1,241–1,945). Had entrants chosen the three largest verifiers at the incumbents’ share of the same year, the 2024 share would have been 0.577 instead of 0.672. The excess choice of the largest verifiers by entrants is present under both regimes (ETS entrants 0.646 versus 0.582 for ETS incumbents; TMS entrants 0.743 versus 0.682), for both designation types and across emission deciles except the lowest and highest (Supplementary Material S5).

The block transfers concentrated the market not by their number but by their destination. Panel C shows that of the clients of the six cancelled verifiers observed in the following year, 47–100% went to a single successor, and for the largest cancellation (82 clients observed) 71 went to one verifier. The counterfactual paths quantify the consequence and its dependence on the benchmark: had those clients been spread over surviving verifiers in proportion to shares, the 2024 HHI would have been 15 points lower (3% of the 503-point rise); had they

Table 2: Structure of the mandatory verification market, 2013–2024*Panel A. Concentration by registry year*

Year	Entities	Verifiers	HHI, clients	HHI, emissions	CR3	Cancellation exits	HHI cf. prop.	HHI cf. equal
2013	599	21	1274	1693	0.536	0	1283	1258
2014	855	19	1411	1947	0.520	0	1417	1400
2015	853	19	1503	1593	0.544	0	1507	1493
2016	895	19	1497	1458	0.553	0	1498	1488
2017	956	18	1569	1498	0.579	1	1556	1515
2018	1002	17	1676	1762	0.602	1	1624	1562
2019	1043	16	1714	1712	0.613	1	1674	1606
2020	1058	16	1698	1707	0.606	0	1666	1606
2021	1077	15	1739	1989	0.628	1	1717	1635
2022	1094	13	1911	2076	0.706	2	1926	1664
2023	1136	13	1780	2300	0.677	0	1784	1570
2024	1164	13	1777	2121	0.672	0	1762	1568

Panel B. Change in the client share of the three largest verifiers of 2024, 2013–2024, by channel

Channel	Clients	Share points
Entrants' first choice of verifier	+837	+0.191
Entities leaving the registry	-419	-0.075
Voluntary switches (net)	-61	-0.052
Cancellation-linked block transfers (net)	+98	+0.084
Cancellation-linked dispersed transfers (net)	+24	+0.021
Pilot-period dropouts (net)	+1	+0.001
Residual (re-entry after a gap)	+2	+0.002
Total	+482	+0.171

Panel C. Where the client books of cancelled verifiers went

Verifier	Cancelled	Absent from	Book	Observed	Successor	Received	Share
V19	2018-2-28	2017	16	15	V2	7	0.47
V18	2019-2-7	2018	16	16	V1	13	0.81
V16	2019-11-1	2019	11	11	V7	7	0.64
V17	2022-3-1	2021	14	14	V3	14	1.00
V15	2022-10-1	2022	10	8	V14	5	0.62
V4	2022-9-2	2022	93	82	V3	71	0.87

Notes: HHI on a 0–10,000 scale. CR3 is the client share of the three largest verifiers of each year; Panel B fixes the three largest verifiers of 2024. ‘Cancellation exits’ counts verifiers whose last registry year is the previous year, excluding the five pilot-period dropouts of 2013–2014. Counterfactual HHI reassigns every cancellation-linked switch to the surviving verifiers in proportion to their prior-year shares (prop.) or equally, holding those clients there until their next voluntary switch. Panel B decomposes the change in the fixed top-3 share (0.501 to 0.672) into numerator channels and the market-growth (denominator) effect; entrants and leavers are net of their denominator effect. Panel C: ‘Cancelled’ is the NIER cancellation date; ‘Absent from’ the first registry year without the verifier; ‘Book’ its clients in its last year; ‘Observed’ those seen in the registry the next year; ‘Received’ those under the largest successor. Verifiers are coded V1–V24 in order of 2013–2024 client-years.

been spread equally, 209 points lower (42%). In 2022 the two benchmarks even differ in sign (−14 and +247). We therefore report both and do not attribute a share of the rise to the cancellations.

Voluntary switching is common (10.6% of entity-years per year; 9.3% for ETS and 8.2% for TMS entities from 2017) and moves clients away from the largest verifiers, so incumbents’ re-selection is not the source of concentration; nor is the three-year designation cycle, since the voluntary switching rate in renewal seasons (9.4%) equals that in other seasons (9.9%).

4.2 Verifier effects and equivalence bounds

Figure 2 plots the verifier effects on certified intensity and Table 3 reports the joint tests, the equivalence counts, the placebo and the identity check; Box 1 states what the null can and cannot say.

In the fixed-factor group (2,441 entity-years, 304 movers, 16 verifier effects) the joint test does not reject (Wald $p = 0.364$; bootstrap $p = 0.367$), the median MDE is 4.5 log points, 10 of 16 effects pass the $\pm 5\%$ equivalence test and the largest absolute 95% confidence limit among the seven largest verifiers is 7.1 log points. One effect is individually significant at 5% (V4, +3.7, $p = 0.029$), the count expected by chance among 16. The $\pm 2\%$ margin is a question the design can put to only one verifier (V16, MDE 1.9), which does not pass; the count of zero passes at $\pm 2\%$ reflects power, not a rejection. The placebo that redraws destination verifiers from industry-by-year shares gives $p = 0.420$. Within the group, $\ln tCO_2$ on $\ln TJ$ has a slope of 0.908 and R^2 of 0.880: intensity is tightly pinned, but not mechanically, so the null is consistent with the tier rules without distinguishing their effect from low physical variance. On the pre-specified subsample of scope-stable pairs (1,913 entity-years, 13 effects) the joint test again does not reject (bootstrap $p = 0.417$), 10 of 13 pass at $\pm 5\%$ and 3 of 13 at $\pm 2\%$.

A post hoc split to the industries with residual standard deviation below 0.15 sharpens the bound: 12 of 12 effects pass at $\pm 5\%$, 8 of 12 at $\pm 2\%$, and the largest absolute confidence

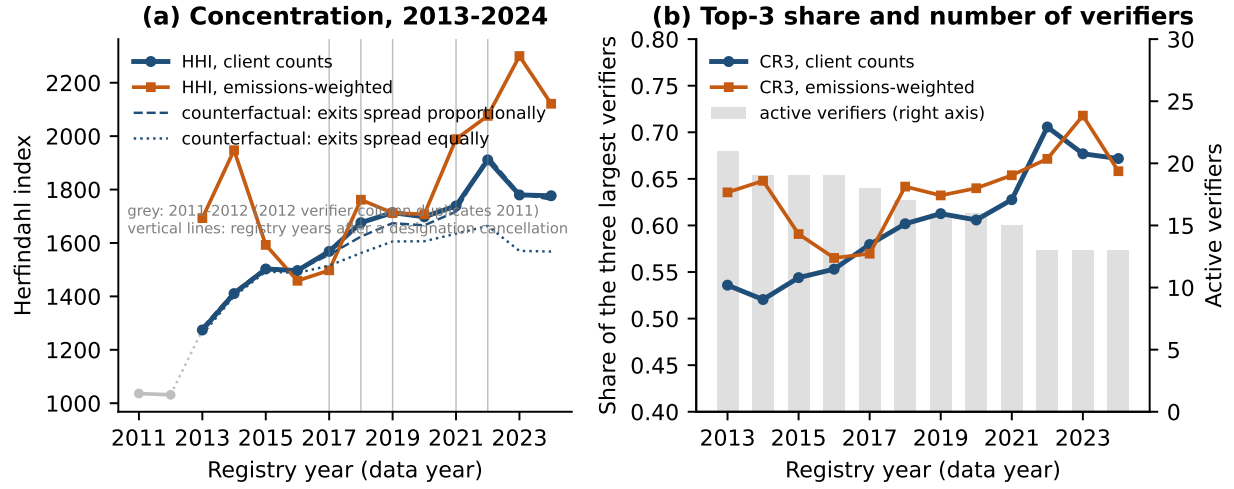


Figure 1: Concentration of the mandatory verification market, 2013–2024. (a) Herfindahl index of client counts and of certified emissions, with the two counterfactual paths of Table 2; vertical lines mark registry years that follow a designation cancellation. (b) Client share of the three largest verifiers (CR3) and number of active verifiers.

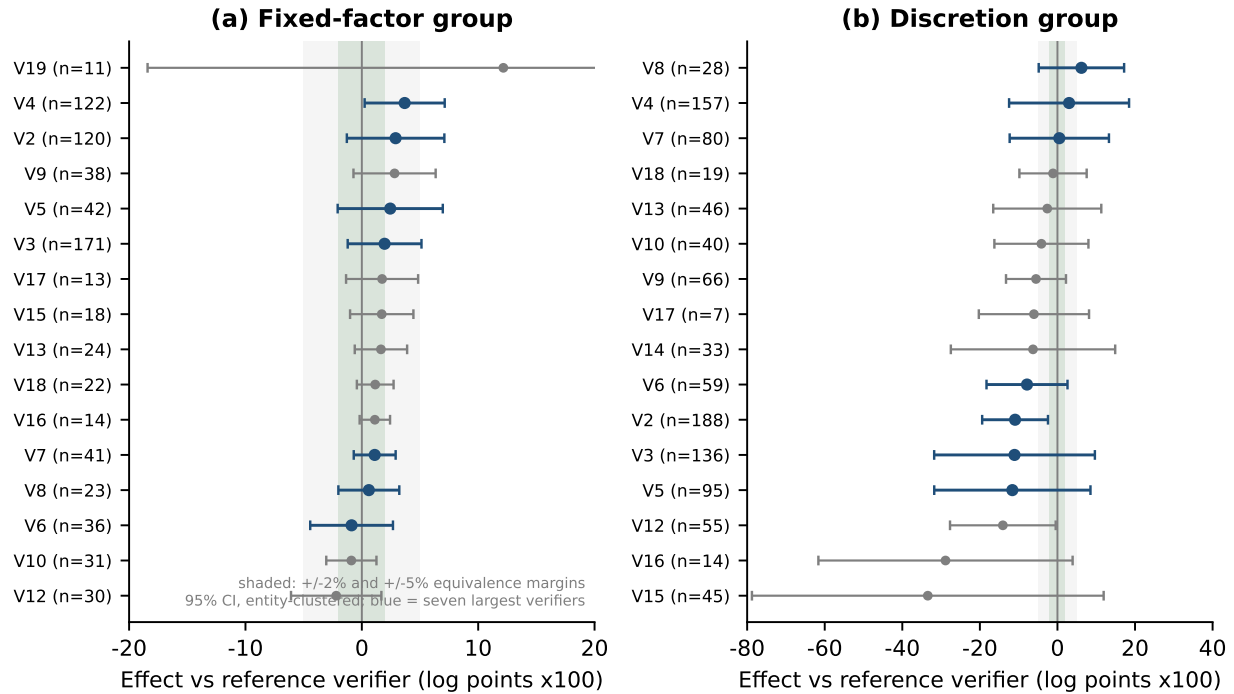


Figure 2: Verifier effects on certified intensity, $\ln(t\text{CO}_2/\text{TJ})$, relative to verifier V1, with 95% confidence intervals (entity-clustered) and the $\pm 2\%$ and $\pm 5\%$ equivalence margins shaded. Entity and industry-by-year fixed effects; movers only; n is the number of switches identifying each effect. (a) Fixed-factor group. (b) Discretion group. Verifiers are coded V1–V24 in order of 2013–2024 client-years; the seven largest are in blue.

Table 3: Verifier effects on certified values: joint tests, equivalence bounds and placebo

Panel A. Joint tests of verifier effects (reference verifier V1)

Sample	Outcome	k	n	Wald p	Boot. p	Holm p	MDE
Fixed-factor group	ln intensity	16	2,441	0.364	0.367	1.000	4.5
	ln emissions	16	2,441	0.005	0.025	0.578	21.9
	ln energy	16	2,441	0.002	0.035	0.739	19.5
Discretion group	ln intensity	16	2,916	0.137	0.241	1.000	19.7
	ln emissions	16	2,916	0.432	0.608	1.000	21.4
	ln energy	16	2,916	0.237	0.407	1.000	22.7
All movers	ln intensity	18	5,449	0.452	0.598	1.000	11.6
	ln emissions	18	5,449	0.190	0.246	1.000	16.3
	ln energy	18	5,449	0.015	0.020	0.482	19.3
Fixed-factor, scope-stable pairs	ln intensity	13	1,913	0.285	0.417	1.000	3.3
	ln emissions	13	1,913	0.042	0.101	1.000	21.1
	ln energy	13	1,913	0.011	0.030	0.663	21.6

Panel B. Equivalence bounds and placebo, ln intensity

Sample	k	Pass $\pm 5\%$	Pass $\pm 2\%$	Judgeable at $\pm 2\%$	Bound	Placebo p
Fixed-factor group	16	10	0	1	7.1	0.420
Discretion group	16	0	0	0	31.8	0.160
All movers	18	1	0	0	18.7	0.535
Low-residual-variance industries (post hoc)	12	12	8	8	3.5	
Fixed-factor, scope-stable pairs	13	10	3	1	5.5	

Notes: Entity and industry-by-year fixed effects; entity-clustered covariance with small-sample correction; wild cluster bootstrap (Rademacher, 999 draws for verifier-level tests, 199 for the joint statistic). Holm adjustment is over the 24 joint tests reported here and in Supplementary Material S6. MDE is the minimum detectable effect at 80% power (2.8 standard errors), median across verifiers, in log points $\times 100$. Equivalence (TOST) is declared when the 90% confidence interval lies inside the margin; ‘Judgeable’ counts verifiers whose MDE is at most 2%. ‘Bound’ is the largest absolute 95% confidence limit among the seven largest verifiers (V2–V8), in log points $\times 100$. Placebo: switch timing kept, destination verifier redrawn from industry-by-year shares (200 draws); p is the share of draws with a larger normalised Wald statistic than the actual one. Identity check: Within the fixed-factor group, ln tCO₂ on ln TJ has slope 0.908 and R^2 0.880 ($n = 5,503$); in the discretion group slope 0.877 and R^2 0.817.

Box 1. What the bounded null on verifier effects can and cannot say

What is estimated	The effect of the verifier of record on certified $\ln(\text{tCO}_2/\text{TJ})$ relative to V1, identified within the 304 entities that switched verifier (2,441 entity-years), conditional on entity and industry-by-year fixed effects, on values that have already passed the regulator’s conformity assessment, in the fixed-factor group (combustion, mobile, building and energy activity classes; entities below 500 ktCO ₂).
What the null says	Joint Wald $p = 0.364$, bootstrap $p = 0.367$, placebo $p = 0.420$; 10 of 16 effects pass $\pm 5\%$ equivalence; the largest absolute 95% limit among the seven largest verifiers is 7.1 log points. That bound is the finding.
What it cannot say at $\pm 2\%$	Only 1 of 16 verifiers (V16) has an MDE of 2% or less, and it does not pass; zero passes at $\pm 2\%$ reflects power. In the post hoc low-residual-variance split, 8 of 12 pass and the bound is 3.5 log points.
Chance findings	1 of 16 effects is individually significant at 5% (V4 +3.7, $p = 0.029$), the number expected by chance.
Where the data are silent	In the discretion group the median MDE is 19.7 log points (range 11.0–64.8); the null there means ‘not detectable below roughly 20%’, not ‘absent’. The fixed-factor group holds 47% of entity-years but 5.4% of certified emissions.
Certified levels	Verifier effects on $\ln$ emissions and $\ln$ energy are jointly significant in the frozen specification (bootstrap $p = 0.025, 0.035$) but not after Holm adjustment (0.578, 0.739) and not once entities whose energy use changed by more than 10% at a switch are excluded (0.64, 0.82; S6).
Mechanism and rules	Within the group $\ln \text{tCO}_2$ on $\ln \text{TJ}$ has slope 0.908 and R^2 0.880: the null is consistent with the tier rules pinning the factor but does not separate that from low physical variance. The activity-class split uses the current Annex 5; earlier versions were not obtained.
Selection and discipline	Movers are observed longer, are more often ETS entities and have more variable intensity than never-movers (S2); the estimates are for movers. Sample rules, split, inference rules, bound definition and reading rules were frozen (SHA-256) before confirmatory estimation; departures are logged (S11).

limit among the major verifiers is 3.5 log points. We report this as a robustness split, not as the headline, because the split was defined after the data were seen.

In the discretion group the joint test does not reject either (bootstrap $p = 0.241$), but the median MDE is 19.7 log points (range 11.0 to 64.8) and no effect passes at $\pm 5\%$: the data are uninformative below roughly 20%. One large verifier is estimated at -10.9 [$-19.4, -2.4$] ($p = 0.011$), one of 16, which we do not interpret. Across all movers (5,449 entity-years, 18 effects) the joint test does not reject and the median MDE is 11.6. The statement is therefore asymmetric: where the rules fix the factor, verifier identity leaves no detectable trace within about ± 7 log points; where they do not, the question is open.

4.3 Cancellation and absorption

The six designation cancellations are the regulator’s most forceful instrument on the intermediary, and the registry shows what happened to certified values when a cancelled verifier’s book moved. Table 4 reports the three block transfers with at least 13 treated entities. In the largest, 71 clients of a verifier that had held 11% of ETS verifications in 2019 appeared under a single successor in 2022. Relative to the successor’s 121 continuing clients, their certified intensity did not move: the average post-transfer coefficient is $+0.5$ log points (s.e. 3.6, bootstrap $p = 0.898$), the leads show no pre-trend ($p = 0.53$ – 0.97), and against all non-switchers of 2022 the estimate ($+0.6$) passes the $\pm 5\%$ equivalence test. Certified emission and energy levels fell by 5–6 log points without approaching significance. The two smaller transfers replicate the intensity result: -0.7 log points (s.e. 0.5; passes $\pm 2\%$) for the 14 clients moved in 2021 and $+0.6$ ($p = 0.67$) for the 13 moved in 2018, although the 2021 event shows a pre-trend in levels ($p = 0.02$) and its level estimates should not be read.

Two readings are excluded by the record. We do not treat these transfers as re-verification by an independent party: no public record states why the designations were cancelled, and the movement of a book to one successor is compatible with the verification team moving with it (auditor rosters were not obtained). Nor do we read the null as a statement about

the cancelled verifiers’ quality. What the registry shows is that the cancellation reallocated the market—one successor received 87% of the largest cancelled book—without a detectable change in the certified number, and that no procedure governing that reallocation exists in public records.

4.4 Exploratory results

Three further results, reported in full in the Supplementary Material, condition the reading. First, verifier effects on certified emission and energy levels in the fixed-factor group are jointly significant in the frozen specification (bootstrap $p = 0.025$ and 0.035) but not after Holm adjustment over the 24 joint tests ($p = 0.578$ and 0.739), not once entity-specific linear trends are added ($p = 0.32$ and 0.39), and not once the 98 movers whose energy use changed by more than 10% at a switch are excluded ($p = 0.64$ and 0.82); the largest single effect is a jump in certified energy at the switch into one small verifier. We read this as an exploratory signal that a change of verifier coincides with changes in activity data and boundaries, not as a verifier effect (S6). Second, in the event study on scope-stable pairs, cancellation-linked switches are preceded by volatility (the coefficient at $t - 1$ is at least as large as at t_0 for every outcome), so switching follows volatility; voluntary switches coincide with a 5.4 percentage-point higher probability of a large revision in certified energy ($p = 0.002$) and no detectable change in intensity (S7). Third, the frozen test of the regulator’s evaluation disclosures finds no detectable change in net client outflow after either disclosure (2019: -0.130 , $p = 0.30$; 2020: -0.041 , $p = 0.32$). On the secondary outcome, the bottom tercile of the 2020 composite score lost clients relative to the top tercile (-0.283 log points, $p = 0.049$); the direction is stable across tercile definitions, a wider control group ($G = 16$, $p = 0.012$) and leave-one-out, and a placebo in the pre-period is clean, but exact randomisation inference over the 252 possible treatment assignments gives $p = 0.079$ and the realised MDE is 0.33 log points. We report it as borderline and low-powered (S9).

Table 4: Certified values after cancellation-linked block transfers: difference-in-differences, ± 3 years

Event (treated)	Control group	Outcome	Post mean (s.e.)	Boot. p	Pre-trend p	TOST $\pm 5\%$
V4 $\rightarrow$ V3, 2022 ($n = 71$)	successor's clients ($n = 121$)	ln intensity	+0.5 (3.6)	0.898	0.529	no
		ln emissions	-5.5 (4.4)	0.210	0.687	no
		ln energy	-6.0 (5.5)	0.277	0.974	no
	all non-switchers ($n = 794$)	ln intensity	+0.6 (2.7)	0.826	0.583	yes
		ln emissions	-5.9 (4.5)	0.198	0.725	no
		ln energy	-6.5 (5.3)	0.216	0.673	no
V17 $\rightarrow$ V3, 2021 ($n = 14$)	successor's clients ($n = 121$)	ln intensity	-0.7 (0.5)	0.210	0.831	yes
		ln emissions	+1.0 (5.0)	0.824	0.019	no
		ln energy	+1.7 (4.9)	0.689	0.015	no
	all non-switchers ($n = 930$)	ln intensity	+0.4 (0.5)	0.479	0.771	yes
		ln emissions	+6.9 (4.6)	0.132	0.012	no
		ln energy	+6.5 (4.5)	0.156	0.009	no
V18 $\rightarrow$ V1, 2018 ($n = 13$)	successor's clients ($n = 273$)	ln intensity	+0.6 (1.4)	0.671	0.741	yes
		ln emissions	+5.3 (4.8)	0.275	0.458	no
		ln energy	+4.6 (4.8)	0.357	0.509	no
	all non-switchers ($n = 797$)	ln intensity	+1.6 (1.6)	0.361	0.561	yes
		ln emissions	-3.6 (4.2)	0.443	0.743	no
		ln energy	-5.2 (4.1)	0.255	0.802	no

Notes: Treated entities are clients of the cancelled verifier that appear under the successor in the transfer year (V4 cancelled 2022-09-02; V17 cancelled 2022-03-01; V18 cancelled 2019-02-07; the register records the administrative decision, which can follow the transfer year). Post mean is the average of the three post-transfer coefficients relative to the year before the transfer, in log points $\times 100$, with entity and industry-by-year fixed effects and entity-clustered standard errors; pre-trend p is a joint Wald test on the three leads. TOST $\pm 5\%$ is passed when the 90% confidence interval of the post mean lies inside ± 5 log points.

5 Discussion and policy implications

The results answer the two questions with which we began, and each answer comes with a bound.

Where concentration arises. If a regulator has reason to manage concentration in its verification market—a question this paper does not settle, since it finds no evidence that the current level is harmful—the arithmetically large channels are not incumbents’ re-selection but two others: the first choice of verifier by newly covered entities, which favoured the largest verifiers in every year and under every regime and would, at the incumbents’ share, have left the 2024 top-3 share at 0.577 rather than 0.672; and the movement of client books after a cancellation, which went to a single successor in 47–100% of cases and for which no procedure exists in public records. The observed pattern of single-successor absorption is a fact about the registry, not a description of a rule. A regulator that wished to act on it would first have to document it. This channel has a close analogue in the audit market, where the clients of a dissolved firm moved largely to its former offices’ acquirers (Kohlbeck et al., 2008; Blouin et al., 2007); the analogy holds for the pattern, not for the cause, which in our case is unrecorded.

Where verifier identity does not matter. Within the activity classes and sizes to which the tier rules apply, the identity of the verifier leaves no detectable trace on certified intensity inside a bound of 7.1 log points for the major verifiers, and inside 3.5 log points in the low-variance industries. Verifier rankings and capability assessments therefore need not target certified intensity in that segment; what they measure there is not the certified number. This is a statement about the rules, not about the verifiers: the null is consistent with the tier rules pinning the factor, as Zeng et al. (2018) argued tier rules would, but the identity check cannot separate that mechanism from low physical variance, and the certified values have already passed the regulator’s conformity assessment. It is also a statement about a segment

that holds 47% of entity-years and 5.4% of certified emissions. The equivalence result does not extend to the discretion segment—process and waste activities and facilities at or above 500 kt, which hold the bulk of emissions—where the median MDE is 19.7 log points. That segment is an evidence gap, not a finding of no effect. Whether supervisory resources should move towards it is a hypothesis this paper does not test; the design reason for expecting verifier judgement to matter there is the same rule that removes it elsewhere.

What a cancellation does. The regulator’s cancellation of a large verifier’s designation reallocated the market without a detectable change in certified intensity. Read with the market-structure result, this says that the instrument’s main effect in this registry was on concentration, not on the number. It does not say that cancellation was unwarranted, because the cause is unrecorded and the transferred clients’ values had passed conformity assessment both before and after.

Exploratory signals. The level effects, the event study and the disclosure test are hypotheses for supervisory design rather than results to act on. The first two are consistent with a change of verifier coinciding with changes in activity data and boundaries rather than in factors; whether that reflects the new verifier’s review or the entity’s own reorganisation cannot be told apart in these data, and the level effects do not survive multiple-testing correction. The disclosure test is consistent with the weak information effects found for penalties in the EU ETS by Calel et al. (2025) and for audit-firm inspections by Lennox and Pittman (2010): the 2019 disclosure of error rates is not detectably associated with client movement, and the 2020 composite score is associated with a loss of clients by the bottom tercile that is borderline under randomisation inference with ten verifiers.

What the evidence does not support. It does not support a ranking of verifiers by the certified values they produce; the verifiers are coded in the estimation tables for that reason. It does not support a cause for the six cancellations, or the claim that the accreditation

transition removed weaker verifiers: the timing coincides for two of them and the register is silent. It does not support a causal effect of evaluation disclosure on the market. It does not support the absence of verifier effects in the discretion segment, or a reallocation of supervision towards it. And it does not support a statement about the accuracy of verification itself, because the design compares verifiers with one another on post-assessment values rather than with an independently measured truth as in Duflo et al. (2013).

The intermediary reading. In the regulator–intermediary–target model (Abbott et al., 2017; Galland, 2017), the regulator’s three instruments on the intermediary did three different things here. The tier rules pinned certified intensity in their segment, so that the intermediary’s identity did not matter; designation and its cancellation shaped the intermediary market, so that a large cancellation concentrated it; and public evaluation moved clients, if at all, weakly. The instruments the regulator controls most directly act on different objects.

External validity. Table 1 shows which features travel. The EU shares the structure that generates our estimates: the emitter selects and pays, verifiers are accredited against the same ISO standards, tier rules fix default factors for small sources, and the regulator can withdraw accreditation. Two differences bound the transfer. The EU has rotated lead auditors since 2021, so a repeated pairing of the kind our estimates are built on is now capped there, and the first rotations fall due after the 2025 data year: the natural test of whether rotation changes certified values is about to become available in the EU registry. China’s arrangement, in which the provincial authority procures the verifiers, removes the selection margin altogether and is the sharper counterfactual for the entry channel we identify. Whether the EU’s accredited-verifier market shows the same entry-driven concentration is a question its registry could answer (Narassimhan et al., 2018).

6 Limitations and conclusion

The activity-class split uses the current Annex 5; earlier versions were not obtained. The certified values are post-conformity-assessment values, which attenuates verifier effects towards zero. The 2012 verifier column duplicates 2011, so the series start in 2013. The cause of the cancellations is not recorded, and auditor rosters were not obtained. Verifier effects are estimated for the 613 entities that switched, which differ from never-movers, and the fixed-factor bound covers a small share of certified emissions. Industry labels for 2020–2024 are bridged from earlier years; a 2013–2019 subsample gives the same null. The paper reports 24 joint tests and many subsample estimates, applies Holm’s adjustment to the joint tests, and rests its disclosure test on ten verifiers.

Within those limits the registry gives a clear picture. Korea’s mandatory verification market concentrated over 2013–2024 through the first choices of newly covered entities and the absorption of cancelled verifiers’ client books by single successors, not through incumbents’ re-selection. Where the monitoring rules fix the emission factor, the verifier’s identity leaves no detectable trace on certified intensity within a bound of about ± 7 log points, and a large cancellation reallocated the market without moving the certified number. Where the rules leave discretion, the data cannot yet say. For the design of MRV systems the lesson is that the rules on factors, not the choice of verifier, pin the certified number in the segments those rules cover, and that the two moments at which the verification market is shaped—entry and cancellation—are the ones for which no procedure exists in public records.

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Use of generative AI A large language model (Claude, Anthropic) was used for code drafting, literature search assistance and language editing under the author’s supervision; all analyses, numbers and interpretations were verified by the author against the frozen analysis files.

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Supplementary Material

S1 panel construction and reconciliation; S2 movers versus never-movers; S3 industry split and Annex 5 transcription; S4 institutional chronology, verifier roster and NIER designation records; S5 switching decomposition and entrants’ choices by group; S6 verifier effects, all samples and outcomes, and the robustness set (Holm adjustment, entity trends, switch-time level jumps, quiet movers, all entities, 2013–2019, cell counts, evaluation–effect rank

correlations, scope-stable bootstrap); S7 switch event study and the ladder from raw to conditional gaps; S8 block-transfer difference-in-differences, all specifications; S9 evaluation disclosures and client reallocation with tercile, randomisation-inference, leave-one-out and placebo checks; S10 [reserved]; S11 frozen analysis plan, feasibility-check statistics and deviation log; S12 reproduction package.

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